Transformer Model for Fraud Detection in Medical Insurance Claims

DISSERTATION REPORT

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BONAFIDE CERTIFICATE

This is to certify that the dissertation report entitled "**Transformer Model for Fraud Detection in Medical Insurance Claims**" submitted by **Aparna A (KH.SC.P2MCA23020)** and **Rillu Mariyam John (KH.SC.P2MCA23044)** for the award of the degree of **Two-Year Master of Computer Applications** under the **Faculty of Computing** is a bonafide record of the work carried out by them under my guidance and supervision at the Department of Computer Science and IT, School of Computing, Amrita Vishwa Vidyapeetham, Kochi Campus.

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DECLARATION

We, **Aparna A (KH.SC.P2MCA23020), and Rillu Mariyam John (KH.SC.P2MCA23044)** hereby declare that the dissertation entitled "**Transformer Model for Fraud Detection in Medical Insurance Claims**" is the record of the original work done by us under the guidance of **Dr. Dhanya R**, Assistant Professor, Department of Computer Science and IT, Amrita Vishwa Vidyapeetham, Kochi Campus. To the best of our knowledge, this work has not formed the basis for the award of any degree/diploma/associate ship/fellowship or a similar award to any candidate in any University.

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ABSTRACT

Fraudulent medical insurance claims pose a significant challenge to healthcare systems and insurance providers, leading to substantial financial losses and inefficiencies. Traditional fraud detection methods, such as rule-based systems and conventional machine learning approaches, often fail to capture evolving fraudulent patterns due to their reliance on predefined rules and limited feature extraction capabilities. In this study, we propose an Anomaly Detection System based on Transformer-based Autoencoders to identify fraudulent claims with high accuracy. Our model employs a self-supervised learning approach where a Transformer-based model learns normal claim patterns and detects anomalies by measuring reconstruction errors. The dataset is preprocessed using label encoding for categorical variables and standard scaling for numerical features to ensure optimal training performance. The system is trained on structured claim data and evaluated using key metrics such as reconstruction error distributions, silhouette scores, and AUC-ROC analysis to distinguish fraudulent claims from legitimate ones effectively. The model is implemented using PyTorch Lightning, with data divided into training (80%) and testing (20%) sets. Anomalies are identified based on a 95th-percentile threshold of reconstruction errors. Our experimental results show that the proposed model significantly improves anomaly detection performance compared to traditional techniques. The Transformer-based system efficiently captures complex dependencies between claim attributes, enhancing fraud detection accuracy. Furthermore, we visualize model performance using confusion matrices, precision-recall curves, and reconstruction error histograms, providing insights into fraud patterns. The system demonstrates scalability and adaptability, making it suitable for real-world deployment in fraud prevention frameworks. Overall, our research highlights the effectiveness of Transformer-based anomaly detection in medical claims fraud detection, offering a robust, scalable, and data-driven solution to combat fraudulent activities in healthcare insurance systems.

INTRODUCTION

Healthcare fraud is one of the simplest and most complex crimes involving false billing, inflated codes, identity theft, and fraudulent claims. It is a growing problem, costing billions of dollars in the insurance sector annually and increasing premiums for policyholders. Fraudulent activities include upcoding, phantom billing, identity theft, duplicate claims, and falsification of medical records, which collectively lead to financial losses, administrative burdens, and compromised patient care. These fraudulent practices not only result in direct monetary losses but also burden healthcare providers with additional oversight and reduce the efficiency of the delivery of medical services. The increasing complexity of fraud tactics and the volume of claims processed daily highlights the need for innovative solutions in fraud detection.

Traditional fraud detection employs rule-based systems that determine whether a claim is fraudulent using pre-configured criteria. These algorithms are effective in detecting established fraud patterns. However, new techniques frequently overlook false alarms, which waste money and knowledge by looking at valid claims, and they are always changing their strategies to avoid detection, making it more difficult for insurers to protect and act on claims data to prevent fraud.

AI-driven fraud detection improves accuracy and reduces costs by automating claim reviews using advanced ML techniques like deep neural networks and SVMs. Hybrid models enhance predictions, while XAI and analytics dashboards support transparency. Challenges include balancing false positives, real-time processing demands, and compliance with privacy laws. Collaboration among insurers, providers, and tech experts is key to ethical, efficient fraud prevention.

RELATED WORKS

Nabrawi et al. conducted a study aimed at developing a health model that autonomously detects fraudulent activities within health insurance claims in Saudi Arabia. The study leveraged supervised machine learning and deep learning techniques, specifically random forest (RF), logistic regression (LR), and artificial neural networks (ANN), to effectively identify fraudulent claims. These methods were selected based on their ability to analyze large datasets and uncover patterns indicative of fraudulent behavior. A key challenge in the study was handling an imbalanced dataset, as fraudulent claims typically represent a small fraction of total claims. To address this issue, the Synthetic Minority Over-sampling Technique (SMOT) was applied, which helps to balance the dataset by generating synthetic examples of the minority class. This step ensures that the models do not become biased toward the majority (non-fraudulent) class and improves their ability to detect fraud. The dataset used in the study was collected from three different healthcare providers in Saudi Arabia, providing a diverse and representative sample of health insurance claims. The study evaluated the performance of the machine learning models using several validation metrics, including accuracy, precision, recall, specificity, F1 score, and the area under the curve (AUC). These metrics provide a comprehensive assessment of how well each model differentiates between fraudulent and legitimate claims. Among the three models tested, ANN outperformed the others, achieving an accuracy of 94.64%. The ANN model demonstrated superior fraud detection capability with a precision of 98.00%, indicating a low rate of false positives. Additionally, it attained a recall of 96.08%, meaning that the majority of fraudulent claims were correctly identified. The F1 score, which balances precision and recall, was 97.03%, further confirming the robustness of the ANN model. Specificity, which measures the ability to correctly classify non-fraudulent claims, stood at 80%, and the AUC was recorded at 88.04%, showcasing the model's overall effectiveness in distinguishing between fraudulent and non-fraudulent cases. The findings of this study highlight the potential of machine learning and deep learning

techniques in automating fraud detection in healthcare insurance claims. The use of ANN, in particular, demonstrates significant promise in accurately identifying fraudulent activities while minimizing false positives. This research contributes to the growing body of knowledge on AI-driven fraud detection and has implications for improving the efficiency and reliability of health insurance systems in Saudi Arabia and beyond.

Hamid et al. investigate the use of pattern recognition and unsupervised learning techniques to detect fraud in health insurance claims. Unlike supervised approaches, which require labeled data, this study employs an unsupervised methodology, making it particularly useful in scenarios where fraudulent cases are not explicitly labeled.

The study integrates Apriori association rule mining with anomaly detection algorithms, enhancing the ability to identify fraudulent activities in health insurance claims. The analysis is conducted using the Centers for Medicare and Medicaid Services (CMS) 2008-2010 DE-SynPUF dataset, which provides synthetic claims data for research purposes. The proposed fraud detection framework follows a two-stage process. In the first stage, Apriori association rule mining is applied to identify frequent patterns within healthcare transactions. This method extracts meaningful rules by analyzing relationships between patients, services, and service providers. The goal is to uncover patterns that may indicate fraudulent behaviors, such as excessive billing for specific procedures or unusual provider-patient interactions. The extracted rules highlight common trends and associations that exist within the dataset.In the second stage, the generated rules serve as inputs for multiple unsupervised anomaly detection models. These models include:

- Isolation Forest (IF) : A tree-based algorithm that isolates anomalies by randomly partitioning data.
- Cluster-based Local Outlier Factor (CBLOF) : A density-based approach that detects anomalies by examining how data points deviate from their respective clusters.

- Empirical Cumulative Distribution-based Outlier Detection (ECOD) : A non- parametric method that models data distributions to identify outliers.
- One-Class Support Vector Machine (OCSVM) : A kernel-based algorithm that learns a decision boundary to separate normal data from anomalies.

The combination of association rule mining with these anomaly detection models enhances the ability to detect fraudulent claims. The methodology successfully identifies 20 potentially fraudulent rules, which provide insights into common fraudulent behaviors in healthcare claims. Notably, the study highlights significant patterns in provider and physician occurrences, as well as anomalies in diagnosis and procedure codes. These findings suggest that fraud is often concentrated among specific providers and that certain medical procedures may be prone to fraudulent claims. One of the key performance metrics used in the study is the silhouette score, which measures the quality of clustering results. Among the tested models, the CBLOF algorithm achieved the highest silhouette score of 0.114, indicating its relative effectiveness in distinguishing fraudulent claims from legitimate ones. The study also emphasizes the critical role of feature engineering in improving fraud detection performance. Selecting relevant features, such as patient demographics, billing patterns, and provider histories, can significantly enhance the accuracy of fraud detection models. Furthermore, the research highlights the limitations of existing healthcare datasets, particularly in terms of data imbalance and the lack of labeled fraud cases, which make unsupervised approaches more suitable for real-world applications.

Overall, the study by Hamid et al. underscores the potential of combining association rule mining with unsupervised learning techniques to detect fraudulent activities in healthcare claims. By leveraging pattern recognition and anomaly detection, this research contributes to the growing body of knowledge in AI-driven fraud detection and presents a scalable approach that can be applied to large and complex healthcare datasets. The findings have important

implications for insurance companies and regulatory bodies seeking to mitigate fraudulent activities and ensure the integrity of healthcare billing systems.

Kumaraswamy et al. [12] explore the potential of Bayesian Belief Networks (BBNs) for healthcare fraud detection, emphasizing their ability to handle uncertainties and challenges associated with imbalanced and incomplete datasets. Unlike traditional machine learning models, which rely on direct feature-to-outcome mappings, BBNs use a probabilistic approach, modeling dependencies between different variables. This makes them particularly suitable for complex environments like healthcare fraud detection, where uncertainty is a key factor. The study focuses on the development and evaluation of a Bayesian Belief Network model using Texas Medicaid outpatient prescription claims data from 2016 and 2017. To assess its effectiveness, the BBN model was compared to traditional machine learning techniques, including logistic regression and random forests. These comparisons were conducted using F-score and area under the curve (AUC) metrics, which are standard measures of classification performance. One of the key findings of the study was that the BBN model performed comparably to traditional machine learning models, demonstrating its viability as an alternative fraud detection approach.

However, when the SMOTE (Synthetic Minority Over-sampling Technique) resampling method was applied to balance the dataset, the BBN model showed a significant improvement in the F-score for the fraud class. This indicates that the model became better at identifying fraudulent claims. However, this improvement came at a cost—there was a slight decrease in the weighted class average F-score, meaning that while fraud detection improved, the model's overall classification balance was slightly affected. The study highlights several advantages of Bayesian Belief Networks over traditional machine learning models. One major benefit is their ability to handle uncertainty, making them robust in scenarios where data is incomplete or ambiguous. Another advantage is their interpretability, as BBNs provide a structured graphical representation of relationships between variables, which makes them more transparent than black-box models like deep learning. Additionally, the study emphasizes the importance of scalability and efficiency, arguing that fraud detection models

should be computationally feasible and easy to implement in real-world systems. The findings reinforce the importance of accurate fraud classification, as false positives can lead to unnecessary audits and financial loss, while false negatives allow fraudulent claims to slip through undetected. The study suggests that BBNs could play a crucial role in healthcare fraud detection, particularly when combined with data resampling techniques like SMOTE to address class imbalances.

Looking ahead, future research could explore hybrid approaches that combine BBNs with deep learning or reinforcement learning to further improve fraud detection accuracy. Additionally, refining feature selection strategies and incorporating diverse datasets from multiple healthcare providers could enhance the model's generalizability and effectiveness. In conclusion, Kumaraswamy et al.'s study highlights Bayesian Belief Networks as a promising alternative to traditional machine learning models for fraud detection in healthcare. By leveraging probabilistic reasoning, BBNs offer a scalable, interpretable, and efficient solution for identifying fraudulent claims, particularly in datasets with missing or imbalanced data. The integration of SMOTE resampling further enhances the model's ability to detect fraud, though with some trade-offs in overall classification balance. As healthcare fraud remains a significant issue, BBNs provide a strong foundation for developing advanced fraud detection models that can be effectively integrated into real-world healthcare and insurance claim monitoring systems.

Surjuse et al. [13] examine the increasing prevalence of healthcare fraud and the limitations of traditional fraud detection methods, emphasizing the need for more advanced and efficient approaches. Traditional methods, such as rule-based systems and manual audits, are often time- consuming, prone to human error, and ineffective in identifying complex fraudulent patterns. These conventional approaches struggle to keep up with evolving fraudulent tactics, making them inadequate for modern fraud detection. To address these challenges, the study explores the role of Artificial Intelligence (AI) in enhancing fraud detection and prevention. AI-powered models, such as Decision

Tree (DT) , Random Forest (RF), and Support Vector Machine (SVM) , have proven to be highly effective in identifying fraudulent claims. These machine learning algorithms can process large volumes of healthcare data, uncover hidden patterns, and detect anomalies that may indicate fraudulent activity. The Decision Tree algorithm is particularly useful due to its interpretability and ease of implementation. It classifies claims based on a series of decision rules, making it a transparent and understandable model for fraud detection. However, Decision Trees can be prone to overfitting, which can reduce their generalization ability when applied to new data. The Random Forest algorithm improves upon this limitation by combining multiple decision trees, thereby enhancing accuracy and reducing overfitting. As an ensemble learning method, Random Forest provides robust fraud detection capabilities by aggregating predictions from multiple trees, making it a reliable approach for identifying fraudulent claims.

The Support Vector Machine (SVM) algorithm is another powerful AI-based fraud detection method. SVM is particularly useful for handling high-dimensional data and complex fraud detection problems, as it creates a decision boundary that effectively separates fraudulent claims from legitimate ones. Its ability to detect subtle and non-linear fraud patterns makes it a valuable tool for healthcare fraud prevention.

In addition to exploring AI-driven methodologies, the research identifies key markers that contribute to an effective fraud detection framework. These markers include feature selection, which ensures that the most relevant variables are used in fraud detection models, and data balancing techniques, which help address class imbalances commonly found in fraud datasets. The study also underscores the importance of real-time fraud detection, as delayed identification of fraudulent claims can lead to substantial financial losses for healthcare providers and insurance companies. In conclusion, Surjuse et al. highlight the growing role of AI in modern fraud detection systems, demonstrating how algorithms like Decision Tree, Random Forest, and SVM significantly enhance fraud prevention. By identifying key markers for effective

fraud detection, the study provides valuable insights into how AI-driven methodologies can be leveraged to combat healthcare fraud more efficiently. As fraudulent schemes continue to evolve, AI-based models offer scalable, adaptive, and data-driven solutions that can strengthen the integrity of healthcare systems and reduce financial losses caused by fraudulent claims.

Settipalli et al. [15] propose the Weighted MultiTree-Density Based Clustering (WMTDBC) model, an innovative unsupervised multivariate analysis approach for detecting fraud in health insurance claims. Unlike traditional fraud detection models that rely on labeled datasets for training, WMTDBC operates in an unsupervised manner, making it highly effective for scenarios where fraudulent cases are not explicitly tagged. This feature is particularly advantageous in healthcare fraud detection, where acquiring labeled fraud data can be challenging due to privacy concerns and the evolving nature of fraudulent schemes. The WMTDBC model processes both categorical and continuous data in two key stages. In the first stage, a Weighted MultiTree (WMT) structure is constructed to analyze healthcare provider profiles, identify relationships between services, and detect anomalies indicative of false or suspicious claims. The WMT framework effectively maps out provider interactions, service patterns, and financial transactions, enabling early detection of potential fraudulent behavior. By leveraging weighted relationships, the model assigns importance to different provider-service interactions, making it more adaptive and responsive to complex fraud patterns. In the second stage, Density-Based Clustering (DBC) techniques are applied to further detect fraudulent claims by analyzing service counts, billing charges, and provider behavior. The model incorporates several advanced density-based clustering algorithms, including:

- Density-Based Spatial Clustering of Applications with Noise (DBSCAN) [15] – Identifies clusters of fraudulent and non-fraudulent claims by analyzing dense regions in the dataset while treating anomalies as noise.
- Ordering Points to Identify the Clustering Structure (OPTICS) [15] – Enhances fraud detection by identifying hierarchical structures in the data and capturing anomalies across varying density levels.

- Density-Based Clustering (DENCLUE) [15] – Detects fraud by modeling data points using density functions and finding regions with significant concentration.
- Generalized Density-Based Spatial Clustering of Applications with Noise (GDBSCAN) [15] – An extension of DBSCAN that allows for more flexibility in detecting fraud across different claim types and provider behaviors.

The WMTDBC model was particularly effective for unlabeled datasets, as it does not require predefined fraud labels for training. This automation of fraud detection without manual labeling enhances both accuracy and efficiency, making it a scalable and practical solution for real-world healthcare fraud prevention. To validate its effectiveness, the WMTDBC model was tested on Centers for Medicare and Medicaid Services (CMS) Part B data, a widely used benchmark dataset for healthcare fraud detection research. The results demonstrated that WMTDBC outperformed existing fraud detection methods, showcasing its superior ability to identify fraudulent claims based on provider service patterns and billing anomalies. The study highlights the significance of unsupervised learning in healthcare fraud detection, particularly when working with large, complex, and evolving datasets. Traditional fraud detection methods often struggle with data imbalance and the need for labeled fraud cases, making them less effective for real-time fraud prevention. In contrast, WMTDBC provides a more adaptive and automated solution, capable of detecting fraud without prior knowledge of fraudulent cases.

In conclusion, Settipalli et al. introduce WMTDBC as an advanced and practical fraud detection model, leveraging unsupervised learning and density-based clustering to identify fraudulent healthcare provider activities with high accuracy. By integrating Weighted MultiTree analysis and multiple density-based clustering techniques, WMTDBC provides a robust, scalable, and automated approach to detecting fraudulent claims in healthcare systems. Its successful application to CMS Part B data demonstrates its potential for real-world implementation, making it a valuable tool for health insurance companies,

regulatory agencies, and healthcare fraud investigators aiming to combat fraudulent billing and service misuse effectively.

Sharifa and Christopher investigate fraud detection in health insurance claims in Kenya, emphasizing the growing prevalence of fraudulent activities and their detrimental impact on economic stability and healthcare service quality. Fraudulent claims not only lead to significant financial losses for insurers and healthcare providers but also reduce the availability and quality of medical services for genuine patients. The study highlights the urgent need for effective fraud detection systems and explores the potential of data mining techniques in addressing this challenge. To develop an efficient fraud detection model, the researchers evaluate several classification algorithms, including Naïve Bayes, Decision Tree (DT), and K-Nearest Neighbour (KNN). These algorithms were chosen for their ability to analyze large datasets, identify hidden patterns, and classify claims as fraudulent or legitimate based on historical data.

- Naïve Bayes (NB) Classifier – A probabilistic model based on Bayes' theorem, which assumes that features are conditionally independent. Despite its simplicity, Naïve Bayes is known for its high efficiency and strong classification performance, particularly in cases where feature dependencies are minimal.
- Decision Tree (DT) – A widely used classification method that organizes data based on a hierarchical structure of decision rules.It is highly interpretable but can sometimes lead to overfitting, especially in complex datasets.
- K-Nearest Neighbour (KNN) – A distance-based classification algorithm that assigns a label to an instance based on the majority class of its nearest neighbors. While effective, KNN is computationally intensive for large datasets and may be sensitive to irrelevant features and data noise.

The experimental results indicate that the Naïve Bayes classifier outperforms the other two algorithms, achieving a 91.79% accuracy rate in fraud detection. This high accuracy demonstrates Naïve Bayes' suitability

for healthcare fraud detection, as it effectively classifies fraudulent claims while maintaining computational efficiency. Based on these findings, the researchers developed a prototype fraud detection system using Naïve Bayes rules. This system automates the fraud identification process, allowing insurance companies and healthcare providers to quickly and accurately detect fraudulent claims.

The study concludes that implementing data mining techniques, particularly Naïve Bayes-based models, can significantly enhance fraud detection efforts. By automating the process, such models reduce financial losses, improve the efficiency of claim verification, and enhance overall healthcare service delivery. The research further emphasizes the need for continuous refinement of fraud detection models, including the integration of more advanced machine learning algorithms and real-time analytics, to keep pace with evolving fraudulent schemes. In summary, Sharifa and Christopher's study provides valuable insights into the application of classification algorithms for health insurance fraud detection in Kenya. By demonstrating the effectiveness of Naïve Bayes in fraud identification, the research paves the way for data-driven solutions to combat fraud, safeguard healthcare resources, and ensure better service delivery in the insurance and healthcare sectors. Future research could focus on expanding the dataset, incorporating additional machine learning models, and developing real-time fraud detection mechanisms to further strengthen fraud prevention strategies.

Massi et al. [19] introduced an innovative two-way unsupervised clustering technique aimed at enhancing the detection of health insurance fraud. This method effectively identifies outliers, thereby reducing the reliance on labeled data, which is often scarce and expensive to obtain. Additionally, the approach addresses challenges related to misclassification and scalability, ensuring a more robust fraud detection system. By leveraging unsupervised learning, the proposed method enables fraud detection without prior knowledge of fraudulent cases, making it highly applicable to diverse healthcare settings. The study focuses on a prevalent fraudulent

practice known as Diagnosis- Related Group (DRG) upcoding, where hospitals deliberately manipulate patient records to secure higher reimbursements. In many healthcare systems, reimbursements are based on DRG classifications, which categorize patients according to the severity of their condition and the expected cost of treatment. When hospitals misclassify patients into higher-paying DRG categories without justification, they increase their financial gains at the expense of insurance providers and government health programs.

Identifying such fraudulent practices is challenging, as the distinctions between legitimate and manipulated classifications can be subtle. To tackle this issue, the authors propose a two-step unsupervised clustering method that relies on administrative healthcare databases. The first step employs k-means clustering to group hospitals based on their treatment behaviors and financial claims. This approach allows the identification of hospitals that exhibit unusual patterns compared to their peers, thereby highlighting potential fraudulent activity. The clustering method groups similar hospitals together while flagging those with significant deviations as potential outliers, warranting further investigation. In the second step, the method supports auditors by analyzing specific fraud-related variables and patient complexity. This deeper analysis helps validate whether the initially detected outlier hospitals are indeed engaging in fraudulent activities.

By considering variables such as patient demographics, medical procedures, and hospital billing practices, auditors can better assess the legitimacy of claims and reduce the risk of false accusations. This additional layer of scrutiny enhances the reliability of fraud detection, ensuring that only truly suspicious cases undergo further investigation. The study tested this methodology on hospital data from Lombardy, Italy, covering the years 2013 to 2015. The model successfully identified ten hospitals exhibiting outlier behavior in terms of DRG classification and billing patterns. While some hospitals required further investigation to

confirm fraudulent activity, the findings demonstrated the effectiveness of the proposed approach. The scalability of the method allows it to be adapted to different diseases, hospital settings, and healthcare systems, making it a valuable tool for fraud detection in various contexts.

Melih Kirlidog and Cuneyt Asuk conducted an important study on fraud detection in health insurance using data mining techniques, with a particular focus on anomaly detection. Fraud in health insurance is a serious issue that occurs when individuals or institutions intentionally deceive insurance providers for financial gain. Such fraudulent activities lead to significant financial losses for insurance companies and government healthcare programs, ultimately increasing the overall cost of healthcare. Detecting fraud is challenging due to the complexity of medical billing, the vast amount of data involved, and the ever-evolving tactics used by fraudsters. To address this problem, the study applied Support Vector Machine (SVM)-based anomaly detection to a Turkish insurance database. The researchers used the SVM algorithm with a linear kernel, which effectively classified records as either normal or anomalous. SVM is a powerful machine learning technique that excels in binary classification problems, making it particularly useful for detecting fraudulent claims. By mapping data into a higher-dimensional space, the algorithm identifies patterns that distinguish legitimate insurance claims from potentially fraudulent ones. The linear kernel was chosen for its efficiency and ability to separate normal and anomalous claims based on key fraud-related features. The research identified several common fraud patterns within health insurance claims. One major pattern was excessive billing, where healthcare providers inflated the cost of services beyond their actual value.

Another significant pattern involved frequent claims made within a short period, which could indicate either patient manipulation of the system or collaboration between healthcare providers and policyholders. Fraudulent collaborations between hospitals and patients were another key issue, where both parties conspired to submit false claims for treatments that

were either unnecessary or never performed. These types of fraudulent activities are difficult to detect using traditional auditing methods, making advanced machine learning techniques like SVM highly valuable. The study demonstrated that the SVM-based anomaly detection approach successfully flagged suspicious claims, supporting fraud investigators in identifying cases that required further scrutiny. By automating the detection process, this method significantly improves the efficiency of fraud investigations, reducing the reliance on manual auditing and minimizing human error. Moreover, by identifying fraudulent claims early, insurance companies can take proactive measures to prevent financial losses and penalize fraudulent activities before they escalate. Beyond detecting fraud, the research highlights the broader impact of using technology in the health insurance sector. By leveraging machine learning and data mining techniques, insurance providers can enhance their fraud detection systems, ultimately leading to lower financial losses. Reducing fraud benefits not only insurance companies but also policyholders, as it helps stabilize and potentially lower insurance premiums for honest customers. As fraudulent claims drive up the overall cost of insurance, effective fraud detection methods contribute to a fairer and more sustainable health insurance system.

Waqas et al. [21] conducted a comprehensive study on insurance fraud detection using machine learning techniques, evaluating the performance of multiple classification algorithms. Insurance fraud is a persistent issue that results in significant financial losses for insurance companies and policyholders. Fraudulent claims can range from exaggerated damages to entirely false reports, making the detection process complex and challenging.

Traditional methods of fraud detection, such as manual audits and rule-based systems, often struggle to keep up with the evolving tactics of fraudsters. To address these challenges, the study investigates the effectiveness of machine learning models in identifying fraudulent claims

more accurately and efficiently. The researchers tested eight different classification algorithms to determine their effectiveness in detecting fraudulent claims. Among the models evaluated were Decision Tree (DT), Support Vector Machine (SVM), Random Forest (RF), Adaboost, and K-Nearest Neighbor (KNN). These algorithms were assessed using key performance metrics, including precision, recall, and F1- score, which provide a balanced evaluation of accuracy and reliability in fraud detection.

The study focused on an auto insurance fraud dataset, a domain where fraudulent claims are particularly prevalent due to factors such as staged accidents, exaggerated vehicle damages, and false injury reports. The findings revealed that the Decision Tree algorithm performed best, achieving an accuracy of 79%. Decision Trees are widely used in classification tasks due to their interpretability and ability to handle complex decision-making processes. The model's strong performance suggests that it effectively captures patterns and anomalies in insurance claims, distinguishing fraudulent activities from legitimate ones. Closely following Decision Tree was Adaboost, a boosting algorithm that enhances classification performance by combining multiple weak learners. Adaboost achieved an accuracy of 78%, demonstrating its effectiveness in refining fraud detection capabilities by focusing on misclassified cases and improving overall model performance. A key takeaway from the research is the crucial role of data preprocessing in improving the accuracy of fraud detection models. Insurance datasets often contain missing values, inconsistencies, and noise, which can negatively impact the performance of machine learning models. The study emphasizes the importance of data cleaning, feature selection, and transformation techniques to enhance model effectiveness. Proper data preprocessing ensures that relevant features are retained, redundant information is eliminated, and the models can learn meaningful patterns without being misled by irrelevant or misleading data points. This research highlights the growing significance of machine learning in the fight against insurance fraud. By leveraging advanced classification algorithms, insurance companies can enhance their ability to detect fraudulent claims efficiently and reduce financial losses.

The study underscores the importance of continuous improvements in fraud detection techniques, particularly through the integration of robust machine learning models and effective data preprocessing strategies. These advancements not only benefit insurers but also create a fairer system for honest policyholders by preventing fraudulent claims from inflating insurance costs.

Pranjal et al. [24] conducted an extensive study evaluating various machine learning models for detecting Medicare fraud, a critical issue that leads to substantial financial losses and compromises the integrity of healthcare systems. Medicare fraud occurs when healthcare providers, such as hospitals, clinics, or individual practitioners, submit false or misleading claims to receive higher reimbursements. Common fraudulent practices include billing for services not rendered, performing unnecessary procedures, and misrepresenting patient diagnoses to claim higher insurance payments. Due to the scale and complexity of Medicare claims data, traditional fraud detection methods struggle to keep up, necessitating the use of advanced machine learning techniques. The study compared multiple machine learning models to determine their effectiveness in identifying fraudulent providers. The researchers assessed each model based on its ability to balance recall and precision— two key performance metrics in fraud detection. Recall measures the ability to correctly identify fraudulent cases, ensuring that suspicious providers are not overlooked, while precision indicates the accuracy of fraud predictions, minimizing false positives. Achieving a strong balance between these metrics is essential, as a high recall with low precision may lead to excessive false alarms, whereas high precision with low recall could result in missed fraudulent activities. Among the models evaluated, LightGBM emerged as the most accurate and effective option. LightGBM, a gradient boosting framework developed by Microsoft, is designed for efficiency and scalability, making it well-suited for large datasets like Medicare claims records. This model performs exceptionally well in classification tasks by leveraging a histogram-based algorithm that reduces memory usage and speeds up training. Its ability to handle imbalanced datasets—where

fraudulent cases are significantly outnumbered by legitimate claims—makes it particularly valuable in fraud detection.

The study found that LightGBM not only outperformed traditional machine learning models such as Decision Trees and Support Vector Machines (SVMs) but also achieved a superior balance between recall and precision. This ensures that fraudulent providers are accurately identified while minimizing the risk of misclassifying legitimate practitioners as fraudulent. By using LightGBM, fraud investigators can prioritize high-risk cases with greater confidence, improving the efficiency of Medicare fraud detection efforts. Overall, the findings highlight the potential of machine learning, particularly LightGBM, in combating Medicare fraud. The study emphasizes the need for continuous advancements in fraud detection techniques to stay ahead of increasingly sophisticated fraudulent schemes. Implementing powerful models like LightGBM can help healthcare systems and insurance providers reduce financial losses, enhance regulatory compliance, and ensure that Medicare funds are allocated to legitimate healthcare services rather than fraudulent activities.

Mohamed et al. introduced an advanced fraud detection system for medical insurance claims, leveraging the power of unsupervised machine learning to identify fraudulent activities at multiple levels. Fraud in medical insurance is a significant concern, as it leads to increased healthcare costs, financial losses for insurance providers, and higher premiums for honest policyholders. Traditional fraud detection methods often rely on labeled data, requiring extensive manual effort for case validation. However, fraud evolves continuously, making it difficult to maintain comprehensive labeled datasets. This study addresses these challenges by implementing a robust anomaly detection system that does not rely on pre-labeled fraudulent cases. The proposed system integrates a majority voting mechanism across multiple unsupervised machine learning algorithms. Specifically, it combines the outputs of 18 different anomaly detection models to enhance the accuracy and reliability of fraud detection. The

ensemble approach ensures that fraudulent claims are identified based on agreement among multiple models, reducing the chances of false positives and improving overall detection performance. By analyzing patterns and irregularities within healthcare claim data, the system effectively detects anomalies that may indicate fraudulent behavior by healthcare providers, doctors, and patients. One of the key strengths of this study is its application to real-world healthcare claim data from Gulf countries. Validating fraud detection models on actual insurance claims allows for a more practical assessment of their effectiveness. The results of the study aligned well with expert evaluations, confirming the system's ability to detect fraudulent activities accurately. Among the different models developed, the patient fraud detection model achieved a precision rate of 79%, demonstrating its capability to accurately distinguish between legitimate and suspicious claims. To ensure the robustness of the fraud detection system, the study employed statistical validation techniques. These methods confirmed that the proposed approach effectively identifies fraudulent cases without excessive false positives. By leveraging unsupervised learning, the system minimizes the need for manual review of claims, significantly reducing the workload for fraud investigators. This not only saves time and resources but also enables insurers to detect and prevent fraud in a more scalable and efficient manner. Overall, the research highlights the immense potential of unsupervised learning in the field of medical insurance fraud detection. By eliminating the dependence on labeled data, this approach allows for continuous and adaptive fraud detection, keeping pace with evolving fraudulent tactics. The integration of multiple anomaly detection models through majority voting enhances the reliability of fraud identification, making it a valuable tool for insurance companies and healthcare regulators. Ultimately, the implementation of such advanced fraud detection systems can lead to reduced financial losses, improved healthcare cost efficiency, and fairer insurance pricing for policyholders.

Rohan et al. conducted an extensive evaluation of 26 deep learning and machine learning models for healthcare fraud detection using the

Ayushman Bharat Pradhan Mantri Jan Arogya Yojana (PM- JAY) dataset. PM-JAY, India's flagship health insurance scheme, provides financial protection to economically disadvantaged families, making it a crucial target for fraud detection efforts. Fraudulent activities in such large-scale public health programs can lead to significant financial losses, misuse of funds, and a decline in the quality of care for legitimate beneficiaries. Given the high volume of claims processed under PM-JAY, detecting fraudulent cases manually is inefficient, necessitating the use of advanced machine learning and deep learning techniques.

The study explored a variety of fraud detection models, including traditional machine learning approaches and advanced deep learning architectures. A key challenge in fraud detection is the imbalance in datasets, where fraudulent claims are significantly fewer than legitimate ones. This imbalance can cause models to favor the majority class (legitimate claims), leading to poor fraud detection performance. To overcome this challenge, the researchers applied various data balancing techniques, including the Synthetic Minority Oversampling Technique (SMOTE), Adaptive Synthetic Minority Oversampling Technique (ADASYN), and Generative Adversarial Networks for Tabular Data (TGANs). These techniques generate synthetic fraudulent samples, ensuring that models learn to recognize fraudulent patterns effectively without being biased toward the majority class. Among the evaluated models, neural networks trained on an undersampled dataset demonstrated the highest performance, achieving an impressive F1 score of 0.95. The F1 score, which balances precision and recall, is a crucial metric in fraud detection, as it ensures that fraudulent claims are identified accurately while minimizing false positives. The success of neural networks in this study highlights the potential of deep learning for handling complex fraud detection tasks, as these models can capture intricate patterns and relationships in healthcare claim data that traditional machine learning algorithms might miss. The study underscores the importance of balancing precision and recall in fraud detection systems. While high precision ensures that flagged fraudulent cases are indeed suspicious, high

recall ensures that most fraudulent activities are detected. Striking the right balance is critical to minimizing false positives, which can lead to unnecessary investigations, and false negatives, which allow fraudulent claims to go undetected. The research findings emphasize the need for continuous improvements in fraud detection methodologies, particularly in refining data preprocessing techniques and model selection strategies.

Additionally, the study suggests further research on applying similar fraud detection techniques to different insurance domains. While the findings were based on healthcare claims under PM-JAY, fraud is a widespread issue in various insurance sectors, including auto, life, and property insurance. Exploring the effectiveness of deep learning models and data balancing techniques in these domains could lead to the development of more robust and generalized fraud detection frameworks. Overall, the research highlights the transformative role of machine learning and deep learning in combating healthcare fraud, paving the way for more accurate, efficient, and scalable fraud detection systems in large-scale insurance programs.

Mingrui Ma et al. present a comprehensive study on transformer-based anomaly detection, offering a fresh perspective on how transformers can be effectively utilized in identifying anomalies across various domains. Transformers, originally developed for natural language processing (NLP), have demonstrated remarkable capabilities in learning complex patterns and dependencies within sequential data. Given their success in NLP, researchers have started exploring their applications in anomaly detection, where detecting rare and unusual patterns is crucial for domains such as healthcare, finance, cybersecurity, and industrial monitoring. The study delves into the fundamental principles of transformers, explaining how their self-attention mechanisms enable models to capture long-range dependencies and contextual relationships within data. Unlike traditional anomaly detection methods, which rely on predefined rules or statistical techniques, transformer-based approaches learn intricate patterns directly from data, making them highly adaptable to evolving anomalies. This

adaptability is particularly valuable in dynamic environments where fraudulent behaviors, cyber threats, or system failures continuously evolve.

A significant contribution of this work is its discussion of the key challenges associated with transformer-based anomaly detection. One major challenge is the computational complexity of transformers, as their self-attention mechanism scales quadratically with input size. This can be particularly problematic when handling large datasets, requiring optimizations such as sparse attention mechanisms or efficient memory management techniques. Another challenge is the interpretability of transformer-based models, as their decision-making processes are often considered black-box. Understanding how these models detect anomalies remains an ongoing area of research. The study also provides a detailed overview of real-world use cases where transformers have shown promise in anomaly detection.

In cybersecurity, transformer models are used to detect network intrusions and malware by identifying unusual traffic patterns. In healthcare, they help uncover fraudulent claims or abnormal patient health records. In industrial settings, transformers are applied to predictive maintenance, identifying early warning signs of equipment failure by analyzing sensor data. These diverse applications highlight the versatility and effectiveness of transformer-based approaches in anomaly detection. To support future research, the paper reviews various datasets commonly used in anomaly detection, ensuring researchers have access to standardized benchmarks for model evaluation. Additionally, it discusses critical evaluation metrics such as precision, recall, F1-score, and area under the receiver operating characteristic (ROC) curve, which are essential for assessing the performance of anomaly detection models. The study also identifies research gaps, including the need for more efficient training techniques, improved interpretability, and domain-specific optimizations for transformer-based anomaly detection.

Finally, the research paper serves as a valuable resource by compiling more than 100 core references on the topic, providing readers with an extensive foundation for further exploration. By outlining future research directions, the study encourages advancements in transformer-based anomaly detection, paving the way for more efficient, scalable, and interpretable solutions. As transformers continue to evolve, their application in anomaly detection is expected to expand, offering improved accuracy and adaptability in identifying anomalies across a wide range of industries.

DATASET DESCRIPTION

According to a recent study published in the US News and World Report, the cost of medical malpractice in the United States is $55.6 billion a year, which constitutes 2.4% of annual healthcare spending. Another 2011 study published in the New England Journal of Medicine revealed that, annually, during the period 1991 to 2005, 7.4% of all physicians licensed in the US had a malpractice claim. These staggering numbers not only contribute to the high cost of healthcare but also result in high premiums for medical malpractice insurance due to the size of successful claims. The financial burden of medical malpractice affects not only healthcare providers but also patients, as it can lead to increased healthcare costs and reduced access to services.

The dataset used in this study contains detailed information about 79,210 malpractice claim payments made over a significant period. It offers insights into the nature and frequency of malpractice claims, helping researchers and policymakers understand trends in medical liability. The dataset encompasses a diverse range of malpractice cases, from minor grievances to severe cases involving loss of life.

The dataset includes the following attributes:

- Amount - The monetary amount of the claim payment in U.S. dollars. The value can range widely based on the severity of the claim and the final settlement.
- Severity - A severity rating assigned to the damage experienced by the patient, ranging from 1 (emotional trauma) to 9 (death). Higher severity cases generally lead to larger claim settlements.
- Age - The age of the claimant in years at the time of the incident. This factor may influence claim settlements and case outcomes.
- Private Attorney - A categorical variable indicating whether the claimant was represented by a private attorney. Legal representation can have a significant impact on claim amounts and success rates.

- Marital Status - The marital status of the claimant, which can be a factor in determining settlement amounts, especially in cases of loss of consortium claims.
- Specialty - The medical specialty of the physician involved in the lawsuit. Certain specialties, such as surgery and obstetrics, may have higher malpractice risks due to the nature of their procedures.
- Insurance - The type of medical insurance carried by the patient, which may influence the likelihood and magnitude of claims.
- Gender - The gender of the patient, can provide insights into trends and biases in malpractice claims.

This dataset serves as a crucial resource for analyzing patterns in medical malpractice claims. By examining claim amounts, severity levels, and demographic factors, researchers can gain valuable insights into the factors contributing to malpractice lawsuits. Additionally, this dataset provides opportunities for anomaly detection, identifying unusual or potentially fraudulent claims that may require further investigation. Healthcare administrators and policymakers can use this information to develop strategies aimed at reducing malpractice occurrences, improving patient safety, and managing insurance costs effectively.

By leveraging data-driven approaches, researchers can also assess how malpractice trends have evolved and how legal and medical interventions impact claim rates. The dataset provides a foundation for predictive modeling, fraud detection, and policy recommendations that can help create a more efficient and just healthcare system.

METHODOLOGY

To effectively analyze medical malpractice claim data and detect anomalies, we propose an anomaly detection framework based on deep learning techniques. Specifically, we employ an Anomaly Transformer model, which is designed to capture intricate patterns within high-dimensional data and identify outliers that deviate from the expected distribution. Our approach consists of data preprocessing, feature representation, anomaly detection using a transformer-based architecture, anomaly scoring, and evaluation.

The dataset is first preprocessed by encoding categorical variables, normalizing numerical features, and splitting the data into training and testing subsets. A linear embedding layer then maps input features into a higher-dimensional space to facilitate learning. The Anomaly Transformer model captures contextual relationships among claim attributes, reconstructs input instances, and identifies deviations from expected values as anomalies. Anomaly scores are calculated using Mean Squared Error (MSE), with a threshold set based on the error distribution in training data. Claims exceeding this threshold are flagged as anomalies. Performance is evaluated using silhouette scores, precision- recall curves, and ROC-AUC scores to ensure robust anomaly detection.

By leveraging deep learning, our method enhances fraud detection accuracy, aiding insurance companies and policymakers in identifying fraudulent claims, reducing financial losses, and optimizing resource allocation within the medical malpractice domain.

Transformer-based Anomaly Detection Model

Transformer, as one of the most advanced neural network models in Natural Language Processing (NLP), exhibits diverse applications in the field of anomaly detection. The model leverages the Transformer architecture to identify anomalies—unusual patterns or outliers—in complex datasets. Unlike traditional models, Transformers utilize a self-attention mechanism to capture long-range dependencies and intricate patterns, making them highly effective for time-series

analysis, cybersecurity, fraud detection, and industrial monitoring. These models detect anomalies by learning normal data distributions and assigning anomaly scores based on deviations from expected patterns. Variants such as the Anomaly Transformer, Time-Series Transformer, and BERT-based models are designed for different anomaly detection tasks, often using reconstruction-based or predictive approaches. Transformers outperform traditional models by handling high-dimensional, noisy data efficiently while being scalable and adaptable across multiple domains.

Advantages:

- Global Feature Extraction: The self-attention mechanism allows the model to analyze relationships across the entire time series, detecting anomalies that traditional local pattern- based methods might miss.

- Flexibility for Complex Data: Transformers can process diverse data types such as multivariate time series, text, and images, making them adaptable to various anomaly detection tasks.

- High Accuracy for Long-Range Dependencies: Unlike RNNs, transformers excel at capturing long-term temporal dependencies, which is crucial for detecting anomalies that develop over extended periods.

- Parallel Processing: The transformer architecture supports efficient parallel computation, significantly speeding up training and inference on large datasets.

Disadvantages:

- Computational Cost: Training large transformer models requires high-performance hardware and substantial computational resources.

- Data Requirements: Transformers often need large datasets to achieve optimal performance, making them less effective in scenarios with limited training data.

- Interpretability Issues: The complex attention mechanism makes it difficult to understand and explain how the model identifies anomalies.

- Potential for Overfitting: Due to their high model capacity, transformers can overfit training data, leading to poor performance on unseen anomalies.

Transformer models excel in anomaly detection across several key areas due to their ability to analyze complex patterns and long-range dependencies. In network security, they detect malicious activity by identifying unusual connection attempts, abnormal traffic volumes, and suspicious payload content in network traffic data. For IoT sensor data, transformers help identify anomalies in industrial settings, such as temperature fluctuations, vibration changes, or unexpected pressure drops, enabling predictive maintenance. In financial transactions, they play a crucial role in fraud detection by analyzing spending patterns over time, and flagging unusual transaction amounts or activities from unexpected locations. Transformers are also highly effective in medical monitoring, where they detect abnormal variations in patient vital signs like heart rate, blood pressure, or EEG signals, potentially signaling health issues. Additionally, in log analysis, they assist in identifying system issues or security breaches by recognizing unusual patterns in log entries, improving cybersecurity and system reliability.

Design

The process begins with data understanding and preparation, involving cleaning to resolve missing values, outliers, and inconsistencies, followed by feature engineering to enhance data transformation. The dataset is then divided into three sets: training, validation, and testing. During model selection and training, the best algorithm is chosen, hyperparameters are optimized, and overfitting is mitigated. The model is evaluated using specific metrics, and an optimal anomaly detection threshold is set. Continuous improvements are made based on evaluation results, aided by visualization tools to analyze errors and key features. Finally, the model is deployed for real-world predictions, with ongoing monitoring to ensure long-term effectiveness.

The main steps in the model include:

- Data Preprocessing
- Anomaly Transformer Model
- Training Process
- Evaluation & Anomaly Detection
    - Detecting Anomalies
    - Metrics & Visualization

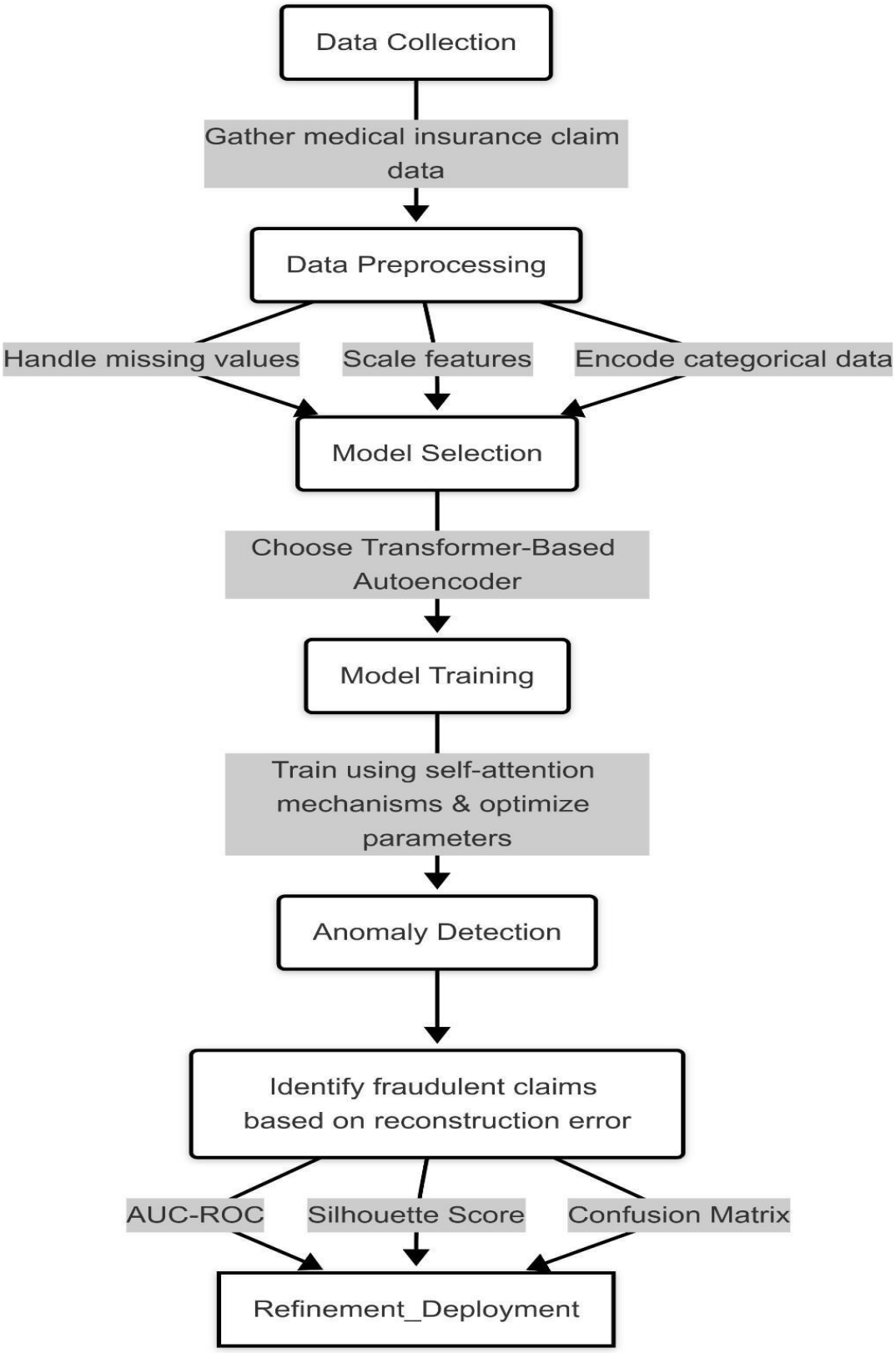

Figure. 1: Workflow of Transformer-Based Anomaly Detection Model

1. Data Understanding and Preparation

The first step in building the anomaly detection model involves understanding and preparing the dataset. The dataset, stored as a CSV file (`medicalmalpractice.csv`), is loaded using `pandas.read_csv()` and examined using `data.info()`. This initial exploration helps in identifying the number of samples, data types, and potential missing values. Understanding the dataset's structure is essential to determine the preprocessing steps required before feeding it into the model. Since machine learning models require numerical data, categorical variables in the dataset—such as `Specialty`, `Insurance`, and `Gender`—are converted into the numeric form using `LabelEncoder()'. Label encoding assigns a unique integer to each category, making the data suitable for model training. For example, `Gender` might be transformed into binary values (`Male = 0`, `Female = 1`), while `Insurance` categories like `Private`, `Public`, and `Uninsured` are assigned integer labels. This ensures that categorical data is efficiently processed by the model.

To ensure that features with different scales do not negatively impact model performance, numerical variables are standardized using `StandardScaler()`. This transformation normalizes the distributions by adjusting them to have a mean of 0 and a standard deviation of 1. Standardization is applied to key continuous variables, including `Amount` (claim amount), `Severity` (case severity), `Age`, `Private Attorney` (legal representation status), and `Marital Status`. By scaling the data, we help the model converge faster and improve anomaly detection accuracy.

The dataset is then divided into training (80%) and testing (20%) subsets using `train_test_split()`. This ensures that the model is trained on a large portion of the data while retaining an independent set for evaluating its performance. The random state is fixed at `42` to maintain consistency and reproducibility across runs. Splitting the data correctly is crucial for ensuring that the model generalizes well to unseen samples.

To make the dataset compatible with PyTorch, a custom `FraudDataset` class is implemented. This class inherits from `torch.utils.data.Dataset` and converts the structured dataset into PyTorch tensors. Each sample can be accessed via the `__getitem__()` method, while `__len__()` returns the total number of data points. By structuring the dataset in this manner, we enable efficient loading and processing during training and evaluation. Finally, the training and testing datasets are wrapped into `DataLoader` objects. The training data loader processes batches of 64 samples at a time, with shuffling enabled to introduce randomness during training. The testing data loader also processes batches of 64 samples, but without shuffling to maintain consistency during evaluation. Using `DataLoader` ensures that the model is trained and tested efficiently, reducing memory usage and improving computational performance.Through these preprocessing steps, the dataset is cleaned, transformed, and structured in a way that optimizes it for anomaly detection. Encoding categorical variables, standardizing numerical features, and properly splitting the data lay a strong foundation for training an accurate and efficient machine learning model.

2. Model Building and Training

The anomaly detection model is built using PyTorch Lightning, a high-level framework that simplifies training deep learning models. The model is designed as an Anomaly Transformer, which leverages a Transformer-based encoder-decoder architecture to learn representations of normal and anomalous data. Transformers are particularly effective for capturing complex patterns and dependencies within sequential or structured data, making them suitable for detecting anomalies in medical malpractice claims.

The Anomaly Transformer model consists of three main components: an embedding layer, a Transformer encoder, and a decoder. The embedding layer projects the input features into a higher-dimensional space, enhancing the model’s ability to learn meaningful representations. The core

of the model is the Transformer encoder, which is composed of multiple `TransformerEncoderLayer` modules. Each layer uses multi-head self-attention to capture dependencies between features and refine the learned representations over several layers. Finally, the decoder maps the transformed embeddings back to the original feature space, reconstructing the input data. The reconstruction error (difference between the input and reconstructed output) is used to identify anomalies.

The model is implemented as a PyTorch Lightning module, which streamlines the training process by handling optimization, logging, and device management. The `forward()` function defines how the input data flows through the model, while the `training_step()` function computes the Mean Squared Error (MSE) Loss, which measures the reconstruction error. The optimizer, `Adam`, is configured with a learning rate of `0.001`, enabling efficient updates to the model's parameters during training.

For training, the model is initialized with the appropriate input dimensions, embedding size, number of attention heads, and Transformer layers. The training loop is executed using `Trainer.fit()`, running for a maximum of 150 epochs. The accelerator is automatically set to `"gpu"` if a GPU is available, ensuring that training runs efficiently on high-performance hardware. Training involves passing batches of data through the model, computing reconstruction loss, and updating the model weights to minimize this error. Over multiple epochs, the model learns to reconstruct normal data patterns, allowing it to differentiate anomalies from normal samples based on reconstruction discrepancies. By leveraging a Transformer-based architecture, self-supervised learning, and reconstruction loss, the model effectively captures the complex relationships in the dataset. The training process ensures that the model generalizes well to unseen data, setting the stage for effective anomaly detection during evaluation.

3. Anomaly Detection and Evaluation

Once the model is trained, the next step is to detect anomalies and evaluate its performance. The anomaly detection process is based on the reconstruction error—the difference between the original input data and the reconstructed output from the model. Since the model has been trained to accurately reconstruct normal data patterns, instances with high reconstruction errors are likely to be anomalies.

To identify anomalies, we first compute reconstruction errors for both the training and test datasets using the `detect_anomalies()` function. This function iterates through the dataset in batches, passes them through the model, and calculates the Mean Squared Error (MSE) for each sample. The reconstruction errors are collected and stored as scores. A threshold is then determined using the 95th percentile of the training reconstruction errors, meaning that any test sample with an error above this threshold is classified as an anomaly.

After detecting anomalies, we proceed with model evaluation. The silhouette score is used as a validation metric to assess how well the model differentiates anomalies from normal cases. A higher silhouette score indicates better clustering, meaning that detected anomalies are distinctly different from normal samples.

To further assess performance, we use standard machine learning metrics, including:

- **Confusion Matrix** – Displays the number of correctly and incorrectly classified anomalies and normal samples.
- **AUC-ROC** (Area Under the Receiver Operating Characteristic Curve) – Measures the model's ability to distinguish between normal and anomalous cases. A higher AUC score indicates better anomaly detection performance.

- **Precision-Recall** (PR) Curve – Evaluates the balance between precision and recall, which is particularly useful when dealing with imbalanced datasets where anomalies are rare.

A histogram is also plotted to visualize the distribution of reconstruction errors in both the training and test datasets. The anomaly threshold is marked on the plot, allowing us to see how many test samples fall beyond the threshold. This helps in understanding the effectiveness of the model in separating normal cases from anomalies.

By leveraging self-supervised learning, reconstruction error analysis, and statistical evaluation metrics, the model successfully detects fraudulent or anomalous claims in medical malpractice data. This evaluation process ensures that the model is not only detecting outliers but also making meaningful and reliable predictions.

4. Refinement

After initial anomaly detection and evaluation, the model undergoes a refinement process to enhance its performance and minimize false positives and false negatives. Refinement is crucial in anomaly detection models since an overly sensitive model may classify too many normal cases as anomalies (high false positive rate), while an overly lenient model may fail to detect actual anomalies (high false negative rate).

One of the primary refinement steps involves adjusting the anomaly threshold. Initially, the threshold is set using the 95th percentile of training reconstruction errors, but this can be fine-tuned based on evaluation metrics. If too many normal cases are flagged as anomalies, the threshold is increased to reduce false positives. Conversely, if the model is missing genuine anomalies, the threshold is lowered to capture more outliers. This balance is fine-tuned based on Precision-Recall AUC and ROC-AUC scores.

Another key refinement step is hyperparameter tuning. The model's performance can be improved by experimenting with:

- Embedding size: Adjusting the feature space representation to capture more meaningful patterns.
- Number of Transformer layers: Increasing or decreasing depth to refine feature extraction.
- Dropout rate: Adding regularization to prevent overfitting.
- Learning rate: Tuning the optimizer's step size to achieve better convergence.

In addition to hyperparameter tuning, feature engineering is explored to improve model inputs. Some variables may have more predictive power in anomaly detection, so feature selection techniques such as correlation analysis, principal component analysis (PCA), or domain knowledge insights are used to refine input features. Removing irrelevant or redundant features helps the model focus on the most critical factors influencing anomalies. Model refinement also involves data augmentation and re-sampling strategies. Since anomalies are rare, techniques such as Synthetic Minority Over-sampling (SMOTE) can be applied to balance the dataset and improve the model's ability to detect outliers effectively. Additionally, different training-validation splits and cross-validation methods are tested to ensure model robustness across various data subsets.

Finally, after implementing refinements, the model is retrained and reevaluated to measure improvements. Metrics such as AUC-ROC, Precision-Recall AUC, and Silhouette Score are analyzed again, and visualizations (histograms, PR curves, and confusion matrices) help confirm the effectiveness of refinements. The goal is to achieve a well-calibrated model that generalizes well to unseen data and minimizes errors in anomaly detection.

5. Visualization and Interpretation

➔ Reconstruction Error Distribution

One of the key visualization techniques used in this workflow is the reconstruction error distribution. This histogram helps in analyzing how well the model reconstructs normal data points compared to anomalies. In an autoencoder-based anomaly detection model, normal data points typically exhibit lower reconstruction errors, whereas anomalies show significantly higher errors. By plotting the distribution for both training and testing data, we can visually assess the range of reconstruction errors and determine a cutoff threshold for detecting anomalies. A clear separation between normal and anomalous data in this plot indicates that the model is performing well, while significant overlap may suggest that further tuning is required.

➔ Confusion Matrix

Another essential visualization tool is the confusion matrix, which provides a detailed breakdown of the model's classification performance. It categorizes the model's predictions into true positives (correctly detected anomalies), true negatives (correctly identified normal cases), false positives (misclassified normal data), and false negatives (missed anomalies). A high false positive rate suggests that the model may be overly sensitive and is flagging too many normal cases as anomalies. Conversely, a high false negative rate indicates that the model fails to detect many actual anomalies, meaning that the threshold might be too high, or the model is not sensitive enough. The confusion matrix plays a vital role in fine-tuning the model's sensitivity and optimizing its decision boundary.

		Predicted Class	
		Positive	Negative
Actual Class	Negative	TP	FN
	Positive	FP	TN

Table 1: Elements of Confusion Matrix

The confusion matrix provides a clear quantitative assessment of the model's strengths and weaknesses. A balanced confusion matrix with low false positives and false negatives, suggests strong performance. However, if the matrix reveals a high number of false negatives, it indicates that the model is missing too many anomalies, which can be critical in high-risk applications. On the other hand, excessive false positives might lead to unnecessary alerts, reducing the model’s reliability in real-world scenarios.

➔ Interpretation of Results

The results of the visualizations provide valuable insights into the model’s effectiveness in detecting anomalies. The reconstruction error distribution helps in setting the right threshold, ensuring that the model correctly identifies anomalous points while minimizing misclassification. If normal and anomalous data overlap significantly in the histogram, the model may require hyperparameter tuning, feature engineering, or additional training data to enhance its ability to distinguish between normal and anomalous cases.

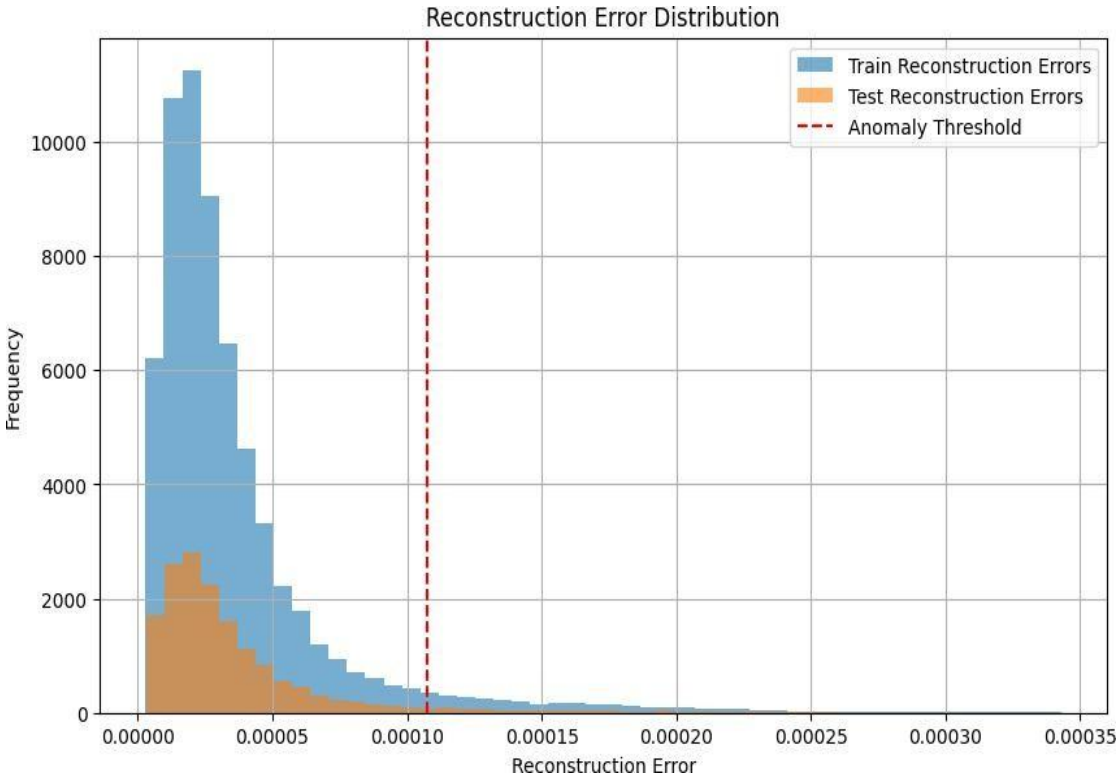

Figure. 2. Distribution of Reconstruction Errors for Anomaly Detection

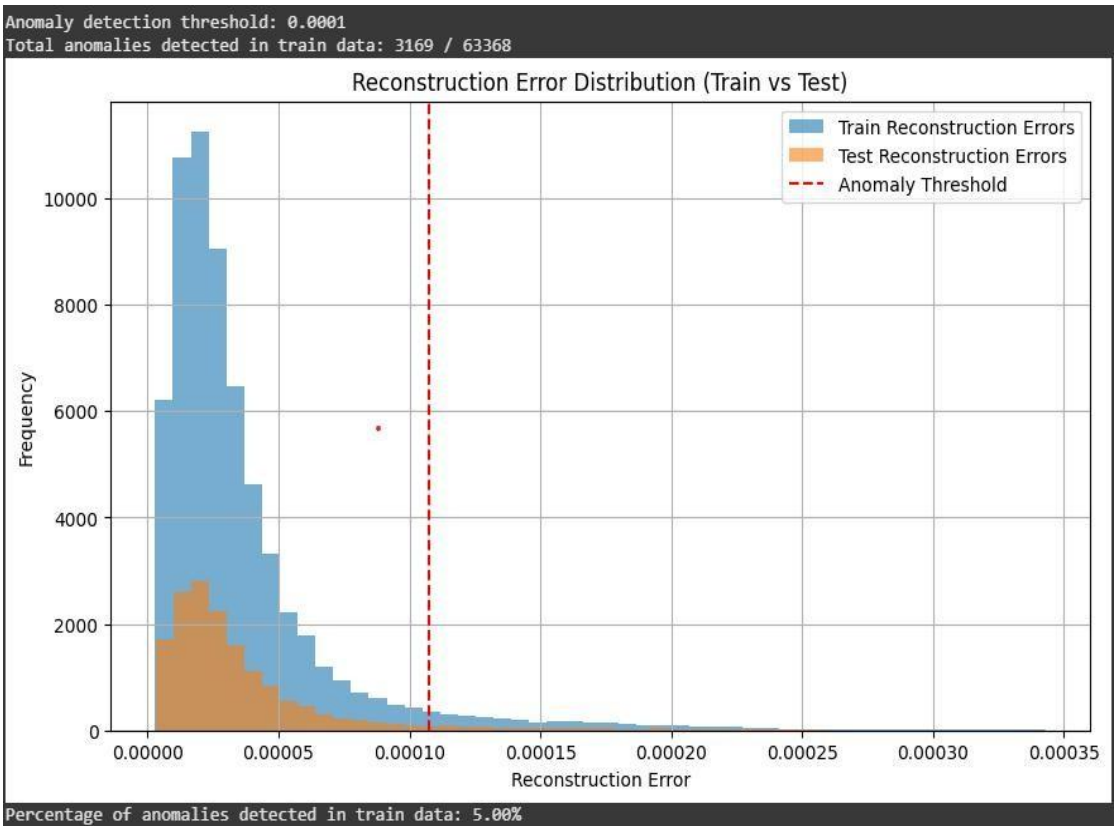

Figure. 3. Reconstruction Error Distribution (Train vs Test)

- Threshold Selection and Anomaly Detection

  Determining the optimal anomaly detection threshold is crucial for the model's effectiveness. If the threshold is set too high, genuine anomalies may go undetected. If it is too low, normal data points might be flagged incorrectly. The reconstruction error distribution helps in visually selecting this threshold, while additional evaluation metrics can further refine the choice. Analyzing detected anomaly indices allows for manual inspection of flagged cases, ensuring that the model is capturing meaningful anomalies rather than noise.

- Performance Metrics for Evaluation

  In addition to visual analysis, quantitative metrics such as the AUC-ROC score provide an objective measure of the model's performance. If ground truth labels are available, the AUC-ROC score helps assess how well the model distinguishes between normal and anomalous data points. A high AUC score (close to 1.0) indicates excellent performance, while a lower score suggests that the model may require further improvements.

  Another useful metric is the Silhouette Score, which evaluates how well anomalies are separated from normal data. A high silhouette score means that detected anomalies form a distinct cluster, while a low score suggests that the anomalies blend in with normal data. If the silhouette score is low, it may indicate that the model needs better feature representation or preprocessing techniques to enhance separation between normal and anomalous cases.

  By combining these visualization techniques and evaluation metrics, we gain a deeper understanding of the anomaly detection model's performance. These insights allow us to refine the model by adjusting the anomaly threshold, optimizing hyperparameters, and improving feature selection. The integration of reconstruction error distribution, confusion matrices, and performance metrics ensures that the model is both effective and reliable for real-world anomaly detection.

RESULTS & DISCUSSIONS

The anomaly detection model achieved a Silhouette Score of 0.32, which suggests a relatively good clustering performance and a clear distinction between anomalous and normal instances. This metric indicates that the model effectively groups similar data points while maintaining a reasonable separation between different clusters, which is crucial for accurately identifying anomalies.

Further supporting this conclusion, the reconstruction error distribution plot demonstrates a noticeable difference in reconstruction errors between normal and anomalous instances. Specifically, the test set exhibits higher reconstruction errors, which aligns with expectations—anomalous data points tend to be more challenging for the model to reconstruct, leading to increased errors. This finding underscores the model's capacity to differentiate between normal and abnormal patterns within the dataset.

Despite these promising indicators, a critical limitation was observed when analyzing the confusion matrix in Fig. 3. The model identified a total of 767 anomalies in the test dataset; however, all instances were ultimately classified as normal. This suggests a potential issue with the model’s sensitivity to rare or extreme cases, leading to false negatives where true anomalies are overlooked. Such a trend can significantly impact the model’s practical usability, especially in applications where anomaly detection is crucial for identifying fraudulent transactions, system failures, or security threats.

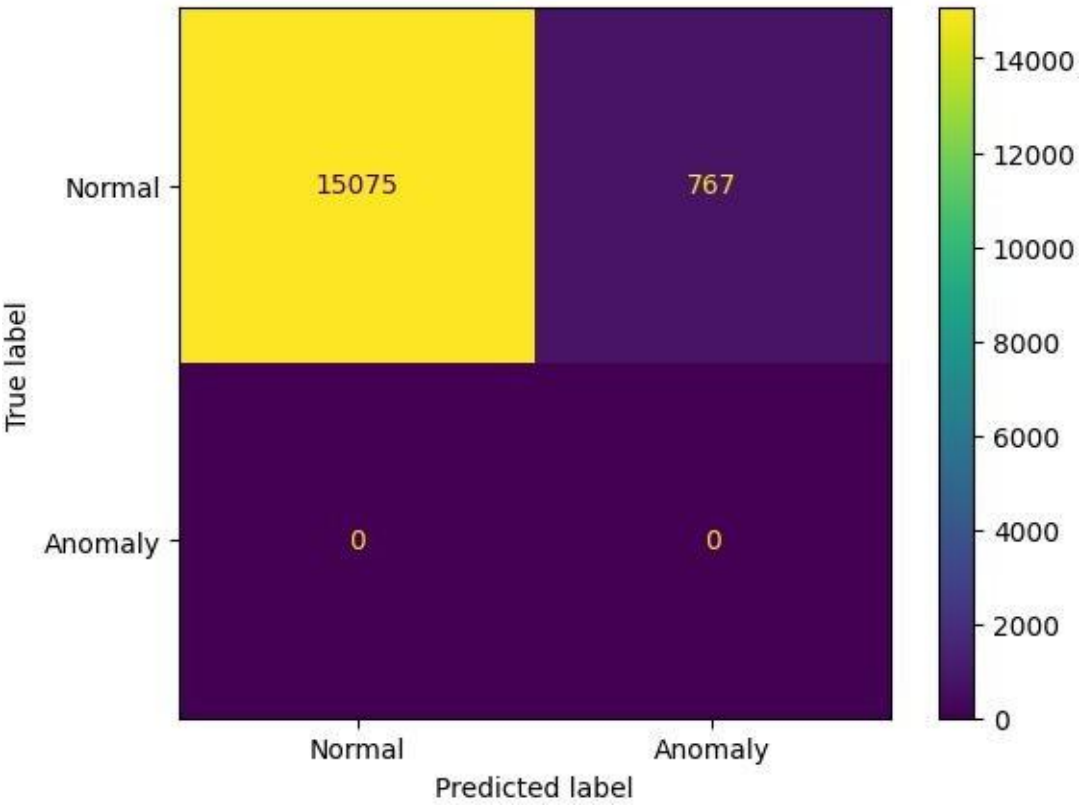

Figure 4. Confusion Matrix for Outlier Detection

CONCLUSION

This study highlights the effectiveness of the Anomaly Transformer model in detecting fraudulent malpractice insurance claims by analyzing complex data patterns. Unlike traditional rule-based approaches, which rely on static heuristics, this deep learning-based method dynamically adapts to evolving fraud tactics. The reconstruction error distribution clearly differentiates between normal and anomalous claims, demonstrating the model's ability to identify irregularities indicative of fraud.

A key advantage of the Anomaly Transformer model is its flexibility and efficiency. By learning from data patterns, it enables real-time fraud detection with minimal manual intervention, reducing reliance on labor-intensive claim reviews. This enhances both accuracy and operational efficiency, allowing insurers to proactively detect and prevent fraudulent claims before they result in financial losses. Additionally, its ability to analyze large datasets provides valuable insights for refining risk assessment and optimizing claims processing.

Beyond the insurance industry, deep learning-based anomaly detection has broad applications in finance, healthcare, cybersecurity, and supply chain management. Its capacity to identify subtle deviations in large datasets makes it a powerful tool for fraud prevention, risk mitigation, and operational optimization.

Ultimately, this study underscores the transformative potential of deep learning in fraud detection. By leveraging advanced anomaly detection techniques, organizations can strengthen security, improve efficiency, and minimize financial risks, ensuring a more resilient and trustworthy system.

FUTURE ENHANCEMENTS

- **Enhance Interpretability with XAI (e.g., SHAP, LIME) -** Fraud detection models often operate as black boxes, making it difficult to understand why a claim is flagged. Using Explainable AI tools like SHAP or LIME allows stakeholders—such as insurers, auditors, and regulators—to see which features influenced a decision. This transparency builds trust, supports regulatory compliance, and enables meaningful human oversight.
- **Integrate Domain Expertise -** Expert-driven heuristics and feature selection help align the model with real-world fraud patterns. Human knowledge can address edge cases or fraud strategies that data alone might miss. This hybrid approach can reduce false positives and improve model credibility in practical deployment.
- **Incorporate External Data Sources -** Using additional datasets like historical fraud reports, external databases, or social network information enhances the model's context-awareness. This enables it to detect more complex, coordinated, or previously unseen fraud schemes that a closed dataset would miss, increasing both recall and precision.
- **Expand to Larger and More Diverse Datasets -** To ensure the model generalizes well, it must be trained and tested on diverse datasets that reflect different fraud types, geographies, and claim categories. This reduces bias and improves the model's robustness when deployed in varied real-world settings.
- **Optimize Anomaly Thresholding -** Setting a rigid threshold may lead to too many false alarms or missed frauds. Adaptive thresholding, possibly guided by claim context, and cost- sensitive learning can balance the trade-off between false positives and false negatives, improving the practical utility of the system.

- **Explore Ensemble Methods -** Different anomaly detection methods (e.g., Transformers, autoencoders, graph models) capture different fraud patterns. Combining them through ensemble techniques enhances overall detection accuracy and resilience, particularly in handling both individual and networked fraud scenarios.
- **Broaden Application Scope (e.g., Risk Assessment, Claims Triage) -** Applying the model to related areas like risk scoring or triage increases its business value and operational efficiency. It helps insurers allocate resources better, prioritize suspicious claims, and improve end-to-end decision-making across the claims lifecycle.

APPENDIX

Transformer Model for Fraud Detection in Medical Insurance Claims

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***Abstract*—Healthcare fraud is a major problem that leads to wasted money and decreased efficiency in medical insurance systems. Traditional fraud-detection methods based on fixed rules struggle to adapt to new scam techniques and often misclassify valid cases as fraudulent. In this article, we put forward a Transformer-Based Anomaly Detection model to enhance fraud detection in medical insurance claims. The model uses self-attention, a feature of transformers, to examine the Medical Malpractice Insurance Dataset and spot unusual patterns by evaluating the reconstruction error thresholds. Developed with PyTorch Lightning, the system's effectiveness is evaluated through metrics that include AUC-ROC, Silhouette Score, and Confusion Matrix. Our results indicate that this model successfully identifies fraudulent claims while significantly lowering the number of false positives. By providing a scalable and adaptable solution, this approach overcomes the limitations of traditional methods, strengthens the fraud detection framework in the insurance industry, and ultimately protects financial and economic efficiency.**

***Index Terms*—Medical Insurance, Healthcare Fraud Detection, Transformer Model, Anomaly Detection, Unsupervised learning**

I. INTRODUCTION

Healthcare fraud is one of the simplest and most complex crimes involving false billing, inflated codes, identity theft, and fraudulent claims. It is a growing problem, costing billions of dollars in the insurance sector annually and increasing premiums for policyholders [1]. Fraudulent activities include upcoding, phantom billing, identity theft, duplicate claims, and falsification of medical records, which collectively lead to financial losses, administrative burden, and compromised patient care [2]. These fraudulent practices not only result in direct monetary losses, but also burden healthcare providers with additional oversight and reduce the efficiency of the delivery of medical services [3]. The increasing complexity of fraud tactics and the volume of claims processed daily highlights the need for innovative solutions in fraud detection.

Traditional fraud detection employs rule-based systems that determine whether a claim is fraudulent using pre-configured criteria. These algorithms are effective in detecting established fraud patterns. However, new techniques frequently overlook false alarms, which waste money and knowledge by looking at valid claims, and they are always changing their strategies to avoid detection, making it more difficult for insurers to protect and act on claims data to prevent fraud.

AI-driven fraud detection, which automates claim reviews, lowers administrative expenses, and improves accuracy, can be advantageous to insurers. Deep neural networks [4] and Support Vector Machines(SVM) [4] are examples of advanced Machine Learning(ML) algorithms that examine large datasets and identify latent fraud tendencies more effectively than conventional techniques. By detecting novel fraud types and improving predictions, hybrid AI models integrate supervised and unsupervised learning, thereby further enhancing fraud detection. Explainable AI(XAI) [5] and dashboards for fraud analytics increase transparency, assisting insurers in making well-informed choices, while preserving patient and healthcare provider trust.

Despite this progress, fraud detection still faces difficulties. Balancing false positives and negatives remains challenging, and real-time analysis requires considerable processing power. While data privacy laws such as compliance with the Health Insurance Portability and Accountability Act (HIPAA) [6] add complexity, stricter detection rules may unfairly reject legitimate claims. Effective and moral fraud detection is necessary to protect patient privacy and the integrity of the healthcare system. Insurers, healthcare providers, and technology specialists should work together to overcome these issues and to build a more open and effective healthcare system.

II. RELATED WORK

Research on insurance fraud detection has primarily focused on fraud committed by hospitals or patients, largely employing data mining and machine-learning techniques. This study takes a more general perspective by investigating all parties to increase their discoveries. It also looks at several fraud-detection systems in other sectors and throughout healthcare, pointing out those with notable results.

Nabrawi et al. [2] explored machine learning for detecting fraud in healthcare claims by focusing on unsupervised models based on limited labeled data. It effectively employs logistic regression [7], Random Forest [8], and neural networks to achieve a high accuracy and strong evaluation metrics.

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